

(Re)Connecting the Dots: Adult Community College Enrollment After State Reconnect Scholarship Programs

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This paper utilizes a quasi-experimental difference-in-difference design that employs a variation in policy timing across states to show changes in adult community college enrollment after the last dollar scholarship program known as "Reconnect" was implemented in Massachusetts, Michigan, and Tennessee. Using geographical proximity and urban to rural population proportions, these states were paired with comparable "control" states New Jersey, Ohio, and Georgia, respectively. Following, a comparative difference-in-differences was run to show descriptive enrollment patterns within these state pairs individually. These samples are then pooled to estimate a combined difference-in-difference model to provide suggestive causal evidence that Reconnect programs can be associated with a 350 student increase in adult community college enrollment. Prior research has been successful in identifying enrollment short term trends after Tennessee implemented a Reconnect program, there is very limited analysis of short term enrollment patterns after implementation in Massachusetts and Michigan due to the novelty of the program, nor does any study examine enrollment increases in a combined multiple state context. This paper highlights the potential that Reconnect programs have in spurring enrollment in the short term, while also emphasizing the need to implement policy that sustains enrollment in the long term.

Keywords: Community College Enrollment, Adult Learners, Reconnect Programs, Last-Dollar Scholarships, Difference-in-Differences, Higher Education Policy, State Financial Aid, Post-secondary Access

I. Introduction and Literature Review

Oftentimes, adults looking to enter or reenter higher education face unique barriers that may discourage enrollment. Understanding these barriers is important when exploring policy options such as last-dollar scholarship programs to get adults back into the classroom. Many adults struggle to return to college due to outside responsibilities, raising concerns about finding time to fit courses into already busy schedules (Acton, 2025). In addition to time constraints, many adults report feeling academically unprepared for new coursework after spending many years away from the classroom. Some adults state that college and work life began to conflict with one another, many saying traditional college schedules

create conflicts with work schedules, as well as long 3–4 month semester-based academic calendars requiring a lengthy disruption to work life balance (Darinestep 2025). The Lumina Foundation found in its 2023 survey of 14,000 students that 85% of respondents reported the cost of attendance as a moderately or very important reason for not currently being enrolled, with work conflicts cited as the second most common barrier (Gallup and Lumina Foundation 2024). Although this study was representative of all participants aged 18–59 years, further research confirmed that adults with an average age of 38 years had specific concerns about the debt from returning to school, among the 29% who cited some sort of barrier stopping them from returning to school (Bellare et al. 2021). These financial barriers are exactly why broader state-level financial aid programs should be accessible to adults seeking higher education.

During the last eight years, some states have aimed to increase enrollment in postsecondary programs for adult learners (“How Community Colleges Can Bring Adults Back to Campus” 2023). This is driven by evidence that the attainment of a post-secondary degree is correlated with higher earning potential. Previous studies have observed that lifetime earnings gains from the attainment of an associate’s degree are around \$450,000 (Carnevale et al. 2024). At the same time, public postsecondary institutions have experienced a decrease in enrollment, with national enrollment levels falling from 617,000 students in spring 2020 to 506,000 in spring 2025 (National Student Clearinghouse Research Center 2025).

Survey data of young and adult community college students reported that the main reason students of all ages decided to drop out of college was due to tuition costs and being unable to finance their degrees (Malhotra and Apicerno 2023). Evidence from Adela Soliz and Coral Flanagan shows the impact that state financial aid programs have on college, applying a similar difference-in-differences model to the one in this paper. This article showed a significant increase in adult enrollment of 8%, 14%, and 10% in West Virginia, Louisiana, and Idaho (Soliz and Flanagan 2025). This article also notes a positive but not statistically significant increase in the enrollment of adult learners in Tennessee (Soliz and Flanagan 2025). These findings provide a basis for further research on the effectiveness of financial aid policies that increase enrollment levels for adult learners.

In 2018, Tennessee launched one of the first Reconnect programs in the United States, which provided a last-dollar scholarship to all qualifying Tennessee adults who do not already have an associate’s or bachelor’s degree and are 23 years or older (State of Tennessee 2025). Immediately after the implementation of the program, the Tennessee Office of Research and Education Accountability noticed an immediate jump in adult enrollment of 24.9% to 28.7% and signaling the policy’s effectiveness (Spiers et al. 2022). Other difference-in-differences (DiD) studies verified this initial jump in enrollment, finding that the program is associated with a difference of 422 students in the two years following its implementation compared to institutions outside of the US (Collom 2022). Although this study highlights the success of the program in the early years, it does not highlight

long-term trends in adult enrollment, which this paper addresses.

Three years later, in February 2021, Michigan announced its own “Michigan Reconnect.” Unlike Tennessee Reconnect, Michigan requires applicants to the program to be 25 years or older (Michigan Reconnect 2025). Preliminary studies on enrollment trends after Michigan Reconnect show an increase in attainment rates from 45% in 2019 to 51.5% in 2022 (Governor Communications Division Office 2024). This signaled program success, however, little follow up was done to confirm that the benefits of this program were sustained.

The most recent Reconnect program, Massachusetts Reconnect, was launched in August 2023. The reason this program is more notable than the others is because it covers all costs related to tuition, fees, books, and supplies in addition to tuition after financial aid (Marceau 2023). The Massachusetts Department of Higher Education reported that 8,411 students ages 25 and older enrolled in community college in the first year of MassReconnect, a 45% increase from the previous year. Although this certainly provides some evidence of a positive relationship between the implementation of the program and the increase in enrollment, more analysis is required to establish whether these trends can be associated with the program itself (Commonwealth of Massachusetts, 2024). Due to the novelty of the program, there is a lack of research on MassReconnect in general (Commonwealth of Massachusetts, 2025).

In this paper, I will utilize a difference-in-differences (DiD) model with institutional and time fixed effects, similar to that of Collom and Soliz and Flanagan, to first perform a descriptive study examining how enrollment patterns change following the introduction of Reconnect programs in Massachusetts, Tennessee, and Michigan. I will do this by pairing the states of Massachusetts, Tennessee, and Michigan with the demographically and geographically comparable states of New Jersey, Georgia, and Ohio. Geographic factors such as urban-to-rural population density, geographic proximity, and overall population size were specifically used to select comparable state pairs.

Urban to rural population ratios were used specifically because proximity to a community college is a large factor in the decision to go to college, as larger distances discouraging enrollment. Increased distance often means increased transportation costs, commute times, and scheduling burdens. Because many adult learners are already constrained by time and financial resources, access to education in this way becomes especially important. More rural states tend to have fewer nearby community colleges, with many students living more than 30 minutes from the nearest institution, placing them in what is known as a “community college desert” (Acton et al. 2025). People in these deserts are far less likely to enroll in college, particularly among low-income and minority populations. Matching states based on these geographic characteristics helps reduce bias from differences in baseline access to higher education, as proximity to a college is strongly linked to enrollment.

Additionally, robustness will be tested using a Comparative Interrupted Time

Series (CITS) design with Newey–West and HAC standard errors to control for autocorrelation and heteroskedasticity. Following this, I will create a pooled sample, in which all treatment and control states are pooled into one sample, and perform a difference-in-differences estimation on just the immediate and year-after effects of the Reconnect policy as a whole.

This paper builds on previous literature in multiple ways. This paper most notably provides the first suggestive evidence of the success of Reconnect programs in a broader context utilizing difference-in-differences and comparative interrupted time series design methods, along with a pooled design, in order to examine the relationship between the implementation of Reconnect and increases in enrollment. Finally, it is the first to examine the comparative effects of Massachusetts Reconnect and Michigan Reconnect, which were newly implemented in 2023 and 2021, respectively. These contributions show how Reconnect policies influence community college enrollment among adults in differing state policy and demographic contexts.

II. Data and Descriptive Statistics

The data used in this analysis were provided by the Integrated Postsecondary Education Data System (IPEDS), the Massachusetts Department of Higher Education, and the New Jersey Department of Higher Education (2024 data). Data include the total enrollment of undergraduate students who are 25 years or older in the Massachusetts/New Jersey sample (National Center for Education Statistics, n.d.[a]) and Michigan/Ohio sample (National Center for Education Statistics, n.d.[b]), and 22 years or older in the Tennessee/Georgia sample (National Center for Education Statistics, n.d.[c]). It is important to note that Tennessee Reconnect was applied to adults 23 years of age and older, instead of 25; therefore, the sample is adjusted accordingly. It is not possible to separate the 22-year-old data from the rest of the sample, so it was necessary to include it for both Tennessee and Georgia, despite Reconnect eligibility beginning at age 23. Because 22-year-olds are included in both treatment and control states and are never eligible for Reconnect, their inclusion does not heavily bias the study but rather creates unnecessary noise in the data. These enrollment numbers represent the number of students who were enrolled during the fall semester of the given academic year (e.g., 2013 represents the fall 2013 semester). Since Reconnect programs went into effect at different times within different states, the time range represented in each DiD estimation may vary slightly. Here I break down the different time frames:

- 1) Tennessee/Georgia DiD: Fall 2013–Fall 2023
- 2) Michigan/Ohio DiD: Fall 2015–Fall 2023
- 3) Massachusetts/New Jersey DiD: Fall 2015–Fall 2024

Using IPEDS, I compiled a list of community colleges located within each state and extract the aforementioned enrollment data at the institutional level. It is important to note that some community colleges had inconsistent enrollment reporting tendencies and these institutions were completely excluded from the study. The enrollment data was then organized by year, institution, and state in which each institution was located. Data from this panel were used in the difference-in-differences model.

The specific states paired with each other in this study were chosen for purpose. Massachusetts was matched with New Jersey due to the similarity in the structure of the state community college systems, similarities in urban and rural areas, and relative geographic proximity. New Jersey and Massachusetts offer undergraduate 2-year degrees at 18 and 15 institutions, respectively. Demographic data was also compared with Massachusetts and New Jersey, which have a fairly similar breakdown, as well as a similar urban and suburban population. Both states have more than 90% of their population located in urban areas, as opposed to rural areas (U.S. Census Bureau 2020; U.S. Census Bureau, n.d.). The same methodology was used to pair Michigan and Ohio. In addition to their geographic location, they are both located in the Rust Belt region, with an urban and rural population breakdown of 71% (MI) and 78% (OH), respectively (U.S. Census Bureau 2020). These two states also have a similar community college structure, with 23 in Ohio and 28 in Michigan, all offering 2-year undergraduate degrees. In the case of Tennessee and Georgia, the comparability standards are also in place, with urban population breakdowns of 73% (GA) and 66% (TN), and rural populations of 27% (GA) and 34% (TN) (U.S. Census Bureau 2020). The number of community colleges in both Tennessee and Georgia are slightly more different than the others, as Tennessee has 13 community colleges and Georgia has 22 community colleges, so this difference should be noted as well for analysis.

Descriptive Statistics for Massachusetts and New Jersey Enrollment

In order to understand the in-depth analysis of adult enrollment, a broader picture must first be painted to show what appeared to be happening to enrollment. Table 1 Panel A presents pre- and post-Reconnect enrollment in both Massachusetts and New Jersey, with pre-policy periods ranging from 2015–2022 and post-policy periods ranging from 2023–2024. Between the pre-policy and post-policy periods, Massachusetts saw an average decline in adult community college enrollment of 15 students, while New Jersey saw a much larger decline of 344 students. Table 1 Panel A helps pinpoint where the average decline was most dramatic, showing that New Jersey enrollment lost an average of 11 students and continued to experience declines thereafter. Massachusetts experienced an increase of 174 students and continued to increase into the fall of 2024 period, where the state had been experiencing overall declines in enrollment in previous years.

Descriptive Statistics for Michigan and Ohio

Table 1 Panel B explores the average adult community college enrollment between Michigan and Ohio institutions during the pre-policy period, spanning from 2015 to 2020 and the post-policy period, spanning from 2021 to 2023. Between Michigan and Ohio, pre-policy enrollment declined by an average of 285 and 1,218 students, respectively. Even considering that Michigan has about half the average enrollment per institution as Ohio does, this highlights a stark decrease in enrollment. It is also evident from Table 4 Panel A that Michigan's enrollment from 2020 to 2021, 2021–2022 and 2022–2023 remained at approximately 1,929–1,932 students, with the state experiencing average declines before 2020. This is not the same in Ohio, as each year after Michigan Reconnect was implemented, Ohio experienced declines in enrollment, except in 2023, where enrollment remained the same. This set of descriptive statistics paints a more confusing picture, as enrollment in Michigan did not necessarily increase after the policy implementation.

Descriptive Statistics for Tennessee and Georgia

Let us now get a sense of enrollment trends in Tennessee and Georgia before and after 2018. Table 1 Panel C outlines the summary statistics of the DiD between Tennessee, which implemented a Reconnect policy, and Georgia, a similar state that did not. Between Tennessee enrollment in the pre- and post-treatment periods, enrollment declined on average by 222 students, while Georgia enrollment declined by 322. This result requires further examination, as enrollment decreases are much greater in Tennessee compared to the other pre- and post averages in Massachusetts and Michigan. This can be done through an examination of the immediate and year-after averages in enrollment. Table 4 Column C in the appendix shows that the average enrollment of Tennessee community colleges increased between the implementation year, 2018 and the year prior. In 2018, the average enrollment for community college students 25 years and older increased by 297 students, from 1,521 to 1,821, and remained at 1,830 in the year following the implementation of Tennessee Reconnect. This improvement in enrollment was short-lived, as in 2020 there was a dramatic decrease in enrollment to 1,571 students. It is important to note that the COVID-19 pandemic was likely the cause of the initial dramatic drop in enrollment; however, enrollment continued on a downward trend thereafter, indicating that enrollment returns may occur only in the short term.

Descriptive Statistics for Pooled Sample

The pooled sample follows a similar pattern to the individual state comparison. Table 1 Panel D shows that the average enrollment of adult community colleges declined much further in states that did not implement Reconnect programs, showing an average decline of 631 students between the pre- and post-periods,

TABLE 1—SUMMARY STATISTICS: MA, NJ, MI, OH, TN, GA

Panel	State	Pre-Mean	Post-Mean	N-Pre	N-Post
A	Massachusetts (Treated)	1772.76	1757.00	96	24
	New Jersey (Control)	1859.26	1515.22	72	18
B	Michigan (Treated)	2215.89	1931.07	108	54
	Ohio (Control)	4731.16	3512.62	48	24
C	Tennessee (Treated)	1728.14	1506.10	55	66
	Georgia (Control)	1628.86	1296.70	80	96
D	Total (Treated)	1905.60	1731.39	259	144
	Total (Control)	2739.76	2108.18	200	138

and within states that did, showing an average decline of only about 174 students, giving us a preliminary look at how Reconnect programs may have influenced enrollment overall. Looking at year-to-year enrollment in Table 3 Column D, the pooled sample beginning five years prior to implementation shows an overall decline in enrollment within both the treatment and control groups. This story changes as enrollment then increases by an average of 125 students leading into the year of implementation and then again by another 116 students in the year following. States that did not implement Reconnect continued to see declines in the year of and after the implementation of Reconnect programs in the treated states. These findings are in line with the short-term findings for each individual state and its comparison states, seeing increases or stabilization in enrollment despite a continued decrease in the control state at the implementation of a Reconnect program.

III. Empirical Strategy

In order to examine differences in enrollment associated with the implementation of the Reconnect program on each individual state level, a DiD model was employed. This method compares the change in undergraduate community college enrollment for students within the target age group before and after the implementation of Reconnect programs in treatment states to the corresponding change over the same time period in similar control states that did not implement a Reconnect program. This model is able to isolate the impact that the Reconnect policy had on enrollment trends from trends that may have occurred regardless of the policy's implementation. The DiD model used for each of the different estimates is as follows:

$$Enrollment_{ijt} = \beta_0 + \beta_1(Post_{jt} \times Reconnect_j) + \mu_i + \lambda_t + \varepsilon_{ijt}$$

The dependent variable, *Enrollment*, represents the number of undergraduate adult community college students enrolled in institution *i* in state *j* in year *t*. The term *Post* is a state-level annual dummy variable that represents whether

or not an observation of institutional-level enrollment i took place before or after the implementation year t of a Reconnect program in state j , where a value of 1 represents that an observation of enrollment did occur in the year during or after the implementation of a Reconnect program, and a value of 0 represents that an observation of enrollment did not occur in the year during or after the implementation of a Reconnect program. The *Reconnect* variable is another dummy that indicates whether state j implemented a Reconnect program, where the value 1 represents that the institution was affected by a Reconnect policy, and where 0 represents that an institution was not impacted by the policy. Finally, coefficient β_1 captures the average difference in enrollment trends after Reconnect, compared to a control state. In order to remove any potential bias from external shocks to enrollment not attributed to the Reconnect policy, the fixed effect λ was added to control for time-variant shocks. Similarly, the fixed effect μ was added to control for any time-invariant unobserved heterogeneity that affects enrollment. The estimates were also tested for parallel trends.

This paper also aims to measure changes in enrollment after implementation of the Reconnect policy by studying the anticipation effects that the announcement of the policy had (Malani and Reif 2015). Enrollment decisions are often forward-looking, so it is appropriate to capture this in the model. In order to capture this in this study, I created the variable *Post_Ant*, which places the treatment year one year before the actual implementation of the policy. The model with anticipation effects is as follows:

$$Enrollment_{ijt} = \beta_0 + \beta_1(Post_Ant_{jt} \times Reconnect_j) + \mu_i + \lambda_t + \varepsilon_{ijt}$$

Unlike the Reconnect programs implemented in Massachusetts and Michigan, Tennessee Reconnect's target age group consists of students 23 years and older. Due to this, it was essential to incorporate data from IPEDS that encapsulate this age group. Age group data was offered in different ranges due to the way data was collected, so age data for students 25 years of age and older were combined with age data for students aged 22–24 years of age, and enrollment was adjusted accordingly. Another important distinction between the Tennessee Reconnect program and Massachusetts Reconnect and Michigan Reconnect is its implementation year. Tennessee Reconnect was implemented in 2018, meaning that to date this program has been running for seven fall semesters. Because of this, it is important to distinguish the immediate post-implementation impacts of the policy from the long-term effects of the policy. To measure the short term, all data after the year 2019 were removed from the study to specifically focus on the year-after impacts on enrollment of the target group. This estimate is in addition to an estimate that captures enrollment trends up to fall 2023.

To estimate enrollment trends after the implementation of Reconnect, our focus coefficient is β_1 . The coefficient β_1 represents the average change in the enrollment in community college undergraduates for the target age group in the states where

the Reconnect policy was implemented, compared to its comparable state, where the Reconnect policy was not implemented, or our counterfactual. Because this study uses panel enrollment data, it is appropriate to utilize institutional-level fixed effects to control for time-invariant characteristics that may impact the coefficient of interest. The presence of institutional-level effects is represented by the symbol μ_i , where i indicates that the effect is on institution i . Additionally, standard errors are institution-robust, but not clustered, as the introduction of clustered standard errors is unreliable due to small group size, which causes severe downward bias.

As an additional robustness check, the enrollment difference between treated and control is analyzed as a single time series with Newey–West HAC standard errors, producing qualitatively similar results.

Pooled Sample Method

Using the sample of institutions from the states listed above, all six states were pooled into one sample and normalized relative to the year that the policy was adopted, as different states adopted the Reconnect policy at different times. Pooling all of the samples increases statistical power and mitigates state-specific shocks. With this data set, it is now possible to estimate the suggestive causal impact of Reconnect policies as a whole on adult enrollment within community colleges in the short term, which means the implementation year and year after implementation. Utilizing a similar difference-in-differences design above with unit and time fixed effects, this paper is able to estimate the average increase in adult community college enrollment. The model is as follows:

$$Enrollment_{ijt} = \beta_0 + \beta_1 Immediate_{jt} + \beta_2 YearAfter_{jt} + \mu_i + \lambda_t + \varepsilon_{ijt}$$

Where *Enrollment* represents enrollment in adult community colleges for institution i in state j in year t . The coefficient β_1 represents the change in enrollment associated with the implementation of Reconnect immediately after the implementation of the policy in state j and time t . The coefficient β_2 represents the change in enrollment associated with the implementation of Reconnect one year after the implementation of the policy in state j in year t . Since implementation of Reconnect policies occurred in different years within each state, time variables were standardized as the number of years before or after policy occurred, with 0 being when the policy was implemented (e.g., 1 = 1 year after implementation, -1 = 1 year before implementation).

This policy is implemented at the state level; therefore, there is a potential for within-group correlation. This means conventional cluster-robust standard errors are unreliable. To correct for this, this study utilizes a wild bootstrap cluster procedure with clustering at the state level, a procedure commonly used when there is not a sufficient number of clusters available for standard robust clustering

(Cameron, Gelbach, and Miller 2007).

IV. Parallel Trends

Parallel trends for each of the samples were tested using three methods: visual analysis of pre-trends, a placebo test in the year that most deviates from the parallel trend, and a joint F-test as a supplementary diagnostic. This paper places less emphasis on the joint F-test than the placebo test, as the limited number of pre-policy years and the existence of anticipation effects make the F-test more sensitive.

Visual analysis of Massachusetts and New Jersey pre-trends shows a relatively parallel decrease in adult community college enrollment before Reconnect's implementation. After running a placebo test, setting the treatment year at 2019 where the trends are most perpendicular, a statistically insignificant p-value of 0.54 is reported, which suggests that there are no significant differences in enrollment trends during that period. Finally, a joint F-test was performed, yielding a p-value of 0.06. This indicates marginal evidence of pre-treatment differences at the 10 percent significance level, though the null cannot be rejected at the conventional 5 percent threshold. This result is expected, as the F-test is more sensitive to small deviations across multiple pre-treatment periods, particularly with a smaller group size.

Parallel trends in Michigan and Ohio visually show that the trends deviate between 2017 and 2018, but remain relatively parallel before that period. These deviations could be explained by the anticipation effects discussed earlier. After running a placebo test, setting the treatment year at 2020 where the trends are most perpendicular, a statistically significant p-value of 0.02 is reported, meaning there is a significant difference in enrollment trends during that period, making verification of parallel trends more uncertain. The joint F-test also rejects parallel trends with a statistically significant p-value of 0.006, suggesting that at least one pre-treatment period violates the parallel trends assumption.

It is visually more difficult to verify parallel trends between Tennessee and Georgia due to the more dramatic decrease in Tennessee's enrollment compared to Georgia, so empirical tests are more necessary in this context. The placebo test, with the treatment year set at 2016, a year before Tennessee Reconnect's announcement, yields a statistically insignificant p-value of 0.55. The joint F-test showed a similar result and supports parallel trends with a p-value of 0.44.

Finally, Figure 4 visually presents the line graph showing enrollment trends in the periods before and after Reconnect implementation for the pooled sample. It appears that treatment and control states decrease at a relatively parallel rate from five years prior to implementation up until year 0, where treatment states begin to deviate. Additionally, the parallel trends assumption was tested empirically by examining the significance of the pre-treatment coefficients using a joint F-test. The results did not show statistically significant differences in the pre-treatment period, with a p-value of 0.14; therefore, the parallel trends assumption

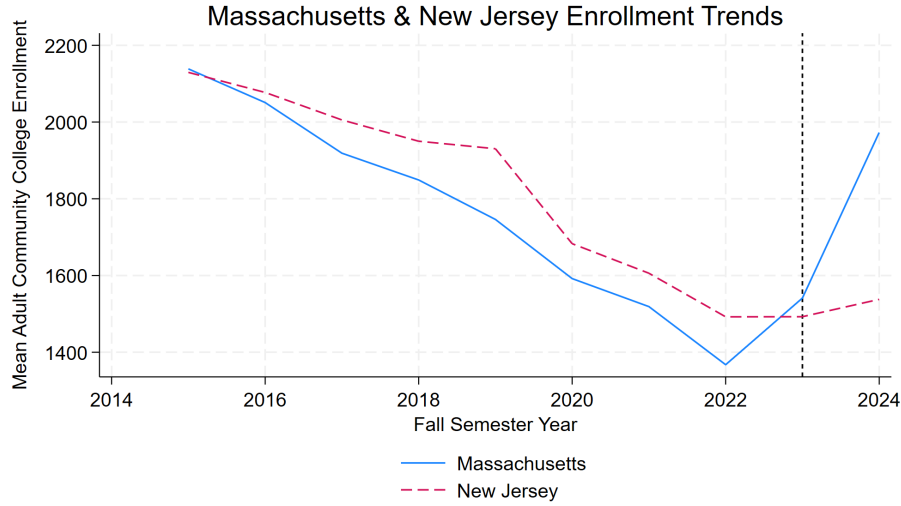


FIGURE 1. MASSACHUSETTS AND NEW JERSEY ADULT COMMUNITY COLLEGE ENROLLMENT TRENDS

was not violated.

V. Results

Massachusetts and New Jersey

Visual changes in enrollment trends by looking at both Massachusetts and New Jersey over time reveal a compelling story. Figure 1 illustrates that, when Massachusetts implemented the Massachusetts Reconnect policy in the fall semester of 2023, the enrollment of adult undergraduate community colleges increased, while New Jersey's continued to decrease. There were no visible jumps in enrollment before Reconnect in Massachusetts or New Jersey. This indicates that enrollment increased only after Reconnect was implemented.

After the visual analysis of the enrollment trends, we can look at Table 2, which reports the results of the DiD estimation. Panel A, Column 1 reports an estimated difference of 328 students relative to New Jersey. For robustness, a CITS was run by comparing the difference between Massachusetts and New Jersey over time, and the results were similar to those shown in the DiD with significance. These results should still be interpreted descriptively due to the inability to cluster. The results of the CITS are shown in Table 4, Panel A in the appendix.

Oftentimes, the effect of a policy can be observed in advance of the actual implementation of the policy itself. Reconnect was officially announced in August 2023 (Commonwealth of Massachusetts 2023), just one month before the official implementation semester of the policy. Panel A, Column 2 reports enrollment

considering anticipation effects. However, the results when anticipation effects are accounted for show a 200-student increase on average. The smaller increase in enrollment when anticipation effects are included suggests that some of the growth in enrollment may have occurred in response to the announcement of the program, rather than the actual implementation of the program. This may indicate that the students adjusted their decisions in advance when they learned that Reconnect would become available in the following academic year.

Michigan and Ohio

In terms of Michigan and Ohio, Figure 2 presents a visual showing the enrollment trends of adult community colleges in Michigan and Ohio before and after Michigan Reconnect. Figure 2 illustrates that, after the implementation year, the average community college enrollment for students 25 years and older in Michigan remained relatively stable, and this stabilization appears to have begun prior to the implementation of Michigan Reconnect, while Ohio's enrollment continued to decrease. This indicates that there might have been another policy action in 2016–2017 that helped stabilize enrollment within Michigan community colleges. The comparison of these two states indicates that Michigan Reconnect may have stabilized enrollment, though it is possible that enrollment had already begun stabilizing due to a prior policy.

Table 2, Panel B, Column 3 reports the difference-in-differences estimate for Michigan and Ohio. The DiD model estimates that, when Michigan implemented its Reconnect program, enrollment in the target group stabilized and is associated with a difference of 934 students relative to Ohio. Panel B, Column 4 incorporates anticipation effects by shifting the implementation year to 2020. The results showed a similar result to that of the DiD, indicating that anticipation probably did not have a major effect on enrollment trends, despite the fact that the program was announced a year earlier (Whitmer 2019). For robustness, a CITS was run by comparing the enrollment difference between Michigan and Ohio over time, and the results were similar to those shown in the DiD with weak significance, further indicating that Reconnect's influence on Michigan's enrollment in adult community college remains somewhat uncertain. The results of the CITS are shown in Table 4, Panel B in the appendix.

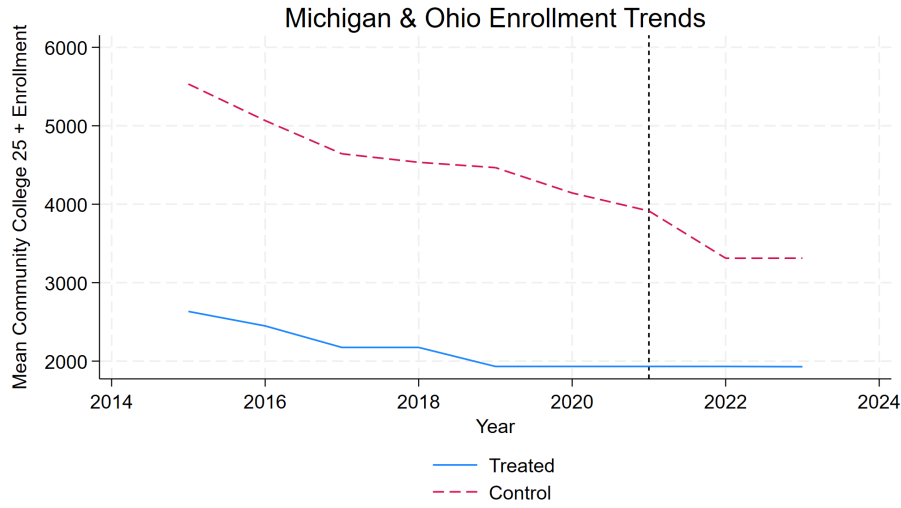


FIGURE 2. MICHIGAN AND OHIO ADULT COMMUNITY COLLEGE ENROLLMENT TRENDS

Tennessee and Georgia

Figure 3 illustrates the difference-in-differences regression line graph for the Tennessee Reconnect target group. Tennessee enrollment follows a generally downward trend until the implementation year of 2018, where there is a dramatic short-term increase in enrollment of the target group, but continued decreases in the two years following the policy. However, prior to this increase in enrollment, there is a smaller jump in enrollment in 2017, which may indicate that there are anticipation effects after the announcement of the policy program prior to its implementation. This would make sense considering that Tennessee Reconnect was announced in 2017 (Gonzales 2018).

Table 2, Panel C, Column 5 reports the results of the difference-in-differences estimation. It appears that when Tennessee implemented its Reconnect program in 2018, enrollment decreased by an average of 63 students in the following periods relative to Georgia. Taking into account the anticipation effects, the result is similar, showing a decrease of 52 students. However, in visual terms, it appears that there was an initial increase in enrollment between 2017 and 2018 and an even greater increase between 2018 and 2019. Isolating only these years and dropping years after 2019 in Column 7 (and Column 8 with anticipation effects), Tennessee saw an increase of 365 adult community college students compared to Georgia. For robustness, a CITS was run by comparing the difference between Tennessee and Georgia over time, and the results were similar to those shown in the DiD with significance. The results of the CITS are shown in Table 4, Panel C in the appendix. These results should still be interpreted descriptively

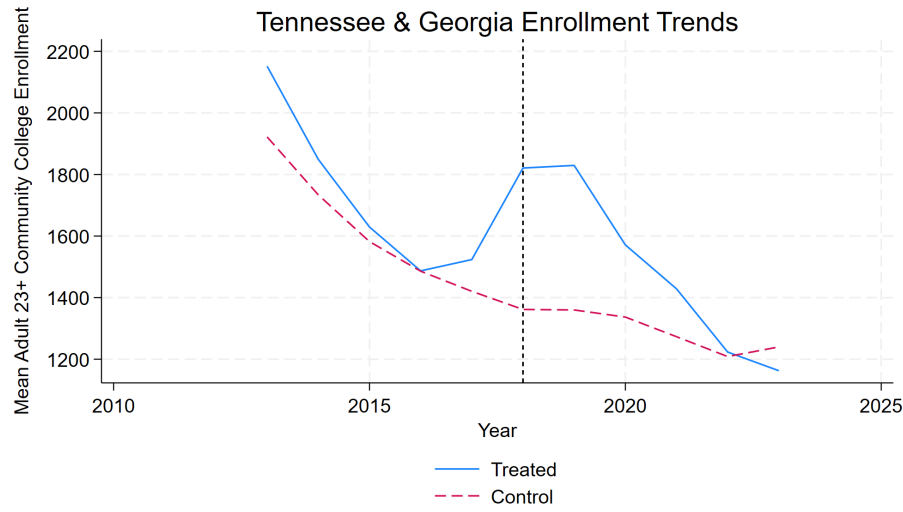


FIGURE 3. TENNESSEE AND GEORGIA ADULT COMMUNITY COLLEGE ENROLLMENT TRENDS

due to the inability to cluster. Taking into account the anticipation effect, the estimated enrollment difference is similar, with an increase of 172 students. This is consistent with the policy announcement in 2017.

Pooled Sample

Visually, according to Figure 4, adult community college enrollment appears to decrease steadily from five years prior to policy implementation to one year prior to policy implementation in both the treatment and control groups. At the time of Reconnect's implementation in states that adopted the policy, enrollment increased in the immediate period and increased even more one year later, while states that did not implement Reconnect continued to experience declines.

Looking at the estimate of the DiD, Table 3, Column A reports the estimated coefficient of both the immediate-term and the year-after term of Reconnect's implementation. It appears that immediately after Reconnect's implementation, adult enrollment in institutions in Massachusetts, Michigan, and Tennessee increased by an average of approximately 164 students compared to adult enrollment in community colleges in Ohio, Georgia, and New Jersey. To confirm significance, the result of the bootstrap analysis produced a statistically insignificant p-value of 0.19 for the immediate period, meaning that it is not certain whether this increase can be attributed to Reconnect. The lack of significance in the immediate period points to potential policy adjustment frictions rather than policy ineffectiveness.

However, looking at the year-after effects of the implementation of Reconnect,

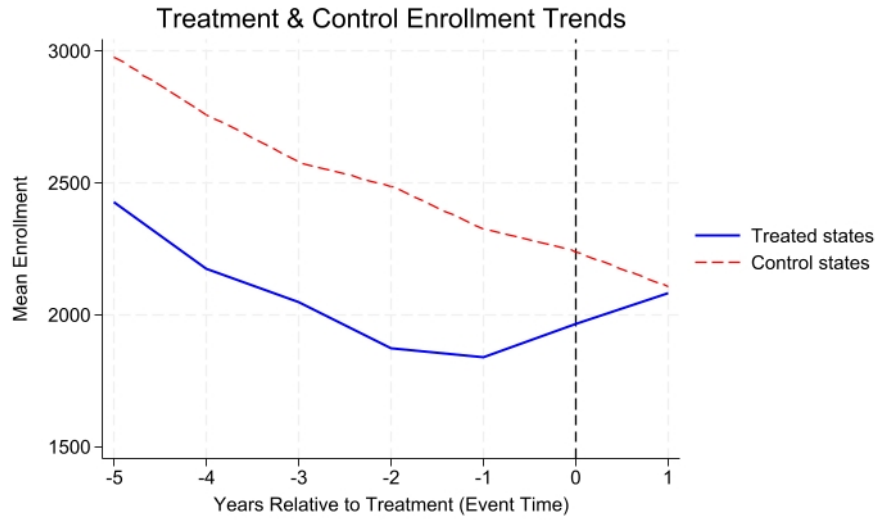


FIGURE 4. POOLED ADULT COMMUNITY COLLEGE ENROLLMENT TRENDS

there is a much more dramatic overall increase in adult community college enrollment compared to states that did not implement Reconnect. Column B reports estimation results and shows that one year after Reconnect, adult community college enrollment increased by an average of 358 students compared to non-Reconnect states. After testing for robustness through the Wild Cluster Bootstrap test, and a resulting p-value of 0.03, the estimate provides stronger evidence consistent with the claim that Reconnect programs likely contributed to enrollment increases in the year after implementation. When performing a bootstrap analysis on both years together, there is a 250-student increase within the two years after implementation of Reconnect policies in the treatment states, with a significant bootstrap p-value of 0.03.

TABLE 2—DIFFERENCE-IN-DIFFERENCES ESTIMATES OF RECONNECT-STYLE PROGRAMS

Adult Community College Enrollment				
Panel A: Massachusetts & New Jersey				
	Fixed Effects	Anticipation Effects		
	(1)	(2)		
Reconnect $\times$ Post	328.28*** (64.58)			
Reconnect $\times$ PostAnt		200.60*** (60.76)		
Observations	210	210		
Institutions	21	21		
Panel B: Michigan & Ohio				
	Fixed Effects	Anticipation Effects		
	(3)	(4)		
Reconnect $\times$ Post	933.7*** (174.92)			
Reconnect $\times$ PostAnt		837.1** (156.6)		
Observations	234	234		
Institutions	26	26		
Panel C: Tennessee & Georgia				
	(5)	(6)	(7)	(8)
Reconnect $\times$ Post	-62.64 (84.85)		365.2*** (72.74)	
Reconnect $\times$ PostAnt		-52.47 (90.84)		171.6*** (90.59)
Observations	297	297	189	162
Institutions	27	27	27	27
Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.				

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 3—POOLED DIFFERENCE-IN-DIFFERENCES RESULTS

	Coefficient (1)	Bootstrap p-Value (1)
Immediate	163.5 (91.92)	0.1875
Year After	357.7 (69.91)	0.0312
Immediate + Year After	250.8 (79.47)	0.0312
Constant	2,247 (54.65)	
Observations	741	
Adj. R^2	0.954	
Number of Clusters	6	

Notes: Robust standard errors in parentheses. Bootstrap p-values reported in the final column. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VI. Limitations

Despite efforts to ensure a robust enrollment estimate after Reconnect, there are still several limitations to this study. Beginning with limitations in the data itself:

first, IPEDS only collects enrollment data in age bins ranging from 22-24 and 25+. This introduces some measurement error as 22 year olds did not receive Reconnect benefits. However, the data do not vary over years, which reduces its impact on DiD comparisons. Second, several institutions had inconsistent or missing enrollment reports and were removed to maintain a balanced panel. Although necessary for model integrity, these exclusions may introduce selection bias if omitted colleges differ meaningfully from those retained. These limitations do not invalidate the descriptive patterns, but they warrant caution when interpreting state-level estimates.

Second, the COVID-19 pandemic impacted all community colleges in the United States, however, differing COVID-19 policy implementations between states may impact enrollment levels around that time. Examining enrollment levels in Massachusetts, there seemed to be a dip in enrollment from 2019 to 2020, around the time the pandemic hit. This, however, was predated by already declining enrollment trends, and no post pandemic recovery until the implementation of MassReconnect when enrollment levels visibly and starkly took an upward trend. A similar event occurred in Tennessee as well when the pandemic hit, the difference was, the pandemic began after the implementation of Tennessee Reconnect. From 2019 to 2020, into 2021, there was a stark dip in enrollment after a plateau from 2018 to 2019. This dip was likely caused by the pandemic but, again, did not stabilize after, signaling that falling enrollment was not solely due to it.

While the potential effects of the pandemic looked similar in Massachusetts and Tennessee, Michigan told a slightly different story. Michigan Reconnect, being implemented in 2021 during the pandemic, was not marked by a decrease in enrollment as Massachusetts and Tennessee were. Michigan saw a stabilization in enrollment levels continuing into 2023, whereas between 2016 and 2019 they were experiencing an overall decline. This may suggest further research must be done in order to understand if the pandemic actually dampened some of the effect that Michigan Reconnect could have had, potentially understating the estimate in this paper.

Comparing the timing of implementation vs the timing of the pandemic is important because of the disruptions the pandemic caused to enrollment, affecting how much the treatment can be isolated. Overall, the setup of this study and the examination of many different states with different policy timing to the pandemic mitigate some of this concern, but this limitation should still be noted. Furthermore, the Reconnect programs differ between the three treated states, which may limit the applicability of these findings to other contexts.

Lastly, the use of a CITS model as an additional robustness check verifies the results of the descriptive study by framing estimates in a different light, while also controlling for potential heteroskedasticity and autocorrelation. However, it does not correct for potential within state institutional correlation, as clustering would, hence, the results are framed descriptively. Additionally, bootstrap clustering within the pooled sample partially mitigates concerns of small cluster inferences,

but statistical power remains limited by the small number of states studied, and results should be interpreted as suggestive causal evidence in the pooled sample only.

VII. Conclusion

This paper finds a positive increase in enrollment trends for adult community college students during the immediate and year after term in both Massachusetts and Tennessee, and pointed to enrollment stabilization in Michigan during the pandemic from both the Difference-in-differences Analysis and the Collapsed Interrupted Time Series Analysis. Massachusetts specifically saw the strongest trend in enrollment and presented the most dramatic shift compared to the other two studies, which may point to Massachusetts' unique approach to cover expenses beyond just tuition.

Tennessee presented a more concerning picture, showing that, after initial and year after increases to adult community college enrollment, it continued on a downward trend thereafter. The state of Tennessee has noted this in previous reports (Tennessee Comptroller of the Treasury 2024). Due to the timing of this decrease, this could have been triggered by the pandemic and the lack of alternate learning platforms, however, this downward trend has persisted up to the most recent enrollment data available, which is fall 2023. This finding confirms what Collom found in his short term Tennessee reconnect study, but adds to this body of research by suggesting that more needs to be done in the long term to maintain enrollment benefits that Reconnect programs may provide community colleges (Collom 2022). This paper was also able to point out potential anticipation effects that suggest enrollment decisions in adult learners may occur at the announcement of the policy, rather than at the time the policy takes effect.

In addition to descriptive trends, the pooled Difference-in-difference study further solidified the findings that the descriptive studies found by allowing inference across different states with varying policies, showing that gains from enrollment are most present in the year following Reconnect implementation. Although the pooled DiD also showed an increase in adult enrollment in the immediate term, the results were statistically insignificant. This evidence suggests that the immediate effects of the policy need to be examined further, but the results of this study suggest that Reconnect made a significant impact on adult enrollment the year after its implementation.

The results of this study provide adequate descriptive and suggestive causal evidence to support the idea that Reconnect policies have a role in spurring enrollment in the short run, but by the trends in enrollment in the longer term seen in Tennessee, may not be adequate in sustaining enrollment. Although these policies have appeared to fulfill their promise in the short term, states need to do more to ensure enrollment does not suffer the same fate as Tennessee. This could look like Massachusetts, where the outside costs of college attendance are considered. Based on the results gathered in this paper, while Reconnect seems to

have some influence on enrollment levels, other policies like expanded availability of night courses, more divided class schedules, or the ability to access online asynchronous programs may need to complement Reconnect programs to sustain enrollment.

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VIII. Appendix

TABLE 4—AVERAGE ADULT ENROLLMENT OVER TIME BY FALL SEMESTER YEAR

Year	Adult Community College Enrollment															
	Michigan		Ohio		Tennessee		Georgia		Massachusetts		New Jersey		Pooled (Treated)		Pooled (Control)	
	μ	SD	μ	SD	μ	SD	μ	SD	μ	SD	μ	SD	μ	SD	μ	SD
2013					2152	1201	1922	875					2139	793	2129	999
2014					1849	1039	1734	770					2051	799	2077	1012
2015	2633	2049	5532	3813	1629	902	1582	699	2139	792	2224	987	2348	1668	3665	3187
2016	2449	1958	5067	3544	1487	767	1486	690	2051	799	2247	981	2429	1645	2978	2242
2017	2175	1797	4643	3368	1524	893	1421	665	1919	719	2141	951	2177	1476	2760	2082
2018	2175	1797	4535	3399	1821	902	1362	587	1849	712	2044	912	2050	1419	2579	2089
2019	1932	1531	4467	3405	1830	800	1360	563	1746	682	2014	899	1875	1223	2488	2089
2020	1932	1531	4144	3129	1517	698	1337	550	1592	650	1742	788	1842	1270	2325	1921
2021	1932	1531	3914	3005	1429	587	1273	572	1519	639	1646	729	1967	1277	2241	1827
2022	1932	1531	3312	2461	1223	489	1209	508	1367	568	1545	629	2083	1246	2108	1491
2023	1929	1500	3312	2505	1163	482	1240	513	1541	629	1534	696	1981	1302	2303	1672
2024									1973	799	1538	787	1874	806	1722	751
<i>N</i>	18		8		11		16		15		11					

Notes: μ denotes mean enrollment, SD denotes standard deviation. Pooled treated group includes Massachusetts, Michigan, and Tennessee,.

TABLE 5—COLLAPSED INTERRUPTED TIME SERIES WITH NEWEY–WEST HAC STANDARD ERRORS

Panel	State	TSinc Coefficient	Post
A	Massachusetts & New Jersey	402.32*** (6.90)	206.77*** (36.63)
B	Michigan & Ohio	190.85* (83.97)	254.68* (102.48)
C	Tennessee & Georgia	-81.19*** (19.35)	515.06*** (73.61)

Note:

Notes: Newey–West HAC standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.